Roll No. | | | | | | | | | | | |

Total No. of Pages :02

Total No. of Questions : 13

B.Pharma (Sem.-6)

BIOPHARMACEUTICS AND PHARMACOKINETICS (THEORY)

Subject Code :BP-604T

M.Code :77989

Date of Examination: 11-12-2025

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains THREE questions carryingTEN marks each and students have to attempt any TWO questions.
3. SECTION-C contains NINE questions carrying FIVE marks each and students have to attempt any SEVEN questions.

SECTION-A

1. **Explainbriefly:**

   a) Why is Vd called 'apparent Vd" and what are its units?

   b) What is tubular reabsorption?

   c) What is the difference between elimination and excretion rate constants?

   d) What are BCS IV drugs?

   e) If the urinary pH becomes too basic, which type of drugs are expected to be retained in the body for long period of time?

   f) Give two examples of therapeutic equivalents.

   g) What is meant by generic substitutes? Give two examples.

   h) What is the relation between total clearance, hepatic clearance and renal clearance?

   i) Write the equation for drug plasma concentration at time't' following IV rapid injection.

   j) How is AUC calculated?

SECTION - B

2. Describe suitable equations to explain Sigma-Minus method and discuss its advantages.

3. Discuss the pharmaceutical factors influencing drug absorption.

4. What is meant by non-linear pharmacokinetics? Discuss the methods used for verifying the non-linear pharmacokinetic behaviour of drugs.

SECTION - C

5. If the amount of drug in the body declines from 100% of the dose (IV bolus injection) to 25% of the dose in 8 hr, what is the elimination half-life of the drug?

6. Two drugs have the following pharmacokinetic parameters after a single oral dose of 500 mg.

DRUG	Ka/hr	k/hr	Vd (mL)
A	1.0	0.2	10,000
B	0.2	1.0	20,000

Both drugs follow one compartment pharmacokinetic model and are 100% bioavailable. Calculate the Tmax for each drug.

7. What is the difference between active and facilitated transport.

8. Distinguish between drug elimination and drug excretion.

9. Is Vd of a drug constant? Comment and mention the conditions under which Vd may change.

10. Briefly explain the method of calculating Ka and Ke (graphically).

11. Draw a typical plasma drug concentration *Vs* time plot and illustrate various pharmacokinetic parameters therein.

12. Discuss BCS of drugs with suitable examples.

13. Discuss with examples various classes of drugs that are exempted from bioequivalence studies.

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NOTE : Disclosure of identity by writing mobile number or making passing request on any page of Answer sheet will lead to UMC against the Student.

NOV 2025